

Stability Analysis and Controller Design of DC Microgrids With Constant Power Loads

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Abstract—DC microgrids consist of multiple power electronic converter units interconnected in a network with sources and loads. They are commonly found in the power systems of electric ships, aircrafts, etc. The main focus of this paper is to study the stability of power converters-based dc microgrids. These high switching frequency electronics are controlled in a way as to maintain constant voltage, current, or power to the load. Due to their high bandwidth, they can be simplified as a constant power load. For this reason, a nonlinear system arises where it is no longer satisfactory to assume local stability. Methods presented in the literature will be reviewed and a method based on semidefinite programming (SDP) will be developed to simplify the stability analysis of these dc microgrids. Furthermore, a robust controller approach based on a similar SDP technique is proposed in order to guarantee and expand the stability region using energy storage.

Index Terms—DC-DC power converters, microgrids, power system control, robust stability, stability analysis.

I. INTRODUCTION

THE RECENT advancement of power electronics presents a wide range of opportunities in the design of microgrids for mission specific applications. In order to increase efficiency and optimize energy management, small/medium voltage dc networks are typically utilized. These dc microgrids are composed of multiple generation units and ac/dc loads interconnected together through filters and power converters. A typical structure of these systems is shown in Fig. 1 [1]–[3].

The controllers of each source or loads in the dc microgrid are designed to track either a voltage, current, or power reference. Due to the advances in power electronics and digital control, these power converters track the desired signal and reject any disturbances with very high accuracy. However, the high bandwidth of the loads introduce a negative impedance effect as seen from the network's perspective. This effect is very commonly seen in tightly controlled loads such as inverter motor drive systems [3], [4], electronic loads, and power supplies [5]. Therefore, typical studies represent these loads as constant power loads (CPLs) [6], [7]. The study of CPLs in dc power networks is fundamental

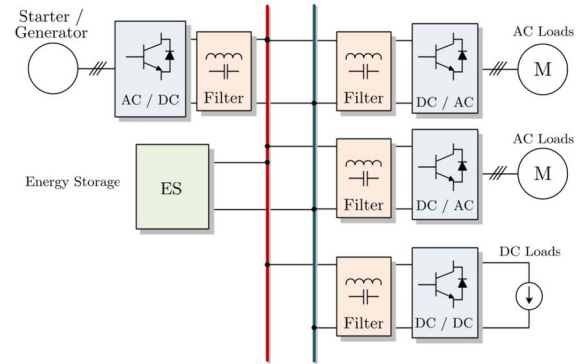


Fig. 1. Detailed dc microgrid of more electric aircraft, navy ships, automobiles, etc.

to automotive [2]–[4], marine systems [8]–[10], and aircraft applications [11], [12].

The design of the filter parameters shown in Fig. 1 is typically carried out locally. For example, the electromagnetic interference filter may be designed based only on its frequency response. However, these filters, as well as the CPLs, will greatly impact the stability and robustness of the dc microgrids. As an example, a smaller capacitor connected to a CPL will decrease the stability margin for the system but a bigger capacitor will increase cost and size [1]. It is therefore important to study the global stability of these systems. One way is to compute a region of attraction (ROA) for an operating point. Different methods of computing an estimation on the ROA have been presented in literature as follows: 1) mixed Lyapunov functions [6]; 2) Takagi–Sugeno (T–S) fuzzy model [1], [2]; 3) genetic algorithms [9]; and 4) reverse trajectory tracking [13].

Among these, the methods based on Mixed Lyapunov functions and Genetic algorithms were shown to be highly conservative while the reverse trajectory tracking method is a graphical approach that is difficult to use in a high-dimensional system. Lastly, the T–S Fuzzy model approach shows good results but its computational complexity grows exponentially with respect to the number of CPLs in the microgrid. It is therefore important to design efficient algorithms to compute the ROA considering overall computation time and memory requirements.

While computing a ROA can be useful for the design of the passive components of the dc microgrid, it is also important to consider the possibility of increasing the robustness of the system. For example, Cespedes *et al.* [5] considered the use of passive damping filters to stabilize the dc network with CPLs.

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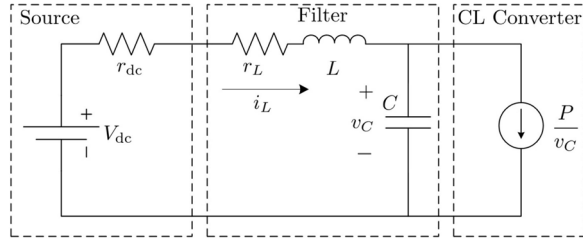


Fig. 2. Simplification of a source and power converter load as a CPL.

However, this method will increase the overall cost and size of the system. A different approach is to use control strategies to mitigate the effects of the CPLs with the use of energy storage or the dc source. One control method considers sliding mode and feedback linearization techniques [3], [14]. However, one drawback is the need to use derivatives which in the presence of noise may not cancel completely the nonlinearity [15]. In [12], a control Lyapunov function is used to generate a feedback law to stabilize the system. The main drawback of this approach is the need to inject a stabilizing current (modifying the load behavior) at all the CPLs. Therefore, this is still an important problem where different approaches can be useful.

This paper focuses on the system shown in Fig. 1. In Section II, a review of the problem is presented based on a simplified case with one CPL. In Section III, a method is proposed to compute the ROA of the dc microgrid with multiple CPLs based on a semidefinite programming (SDP) formulation. Considering the use of energy storage, a robust controller design is presented in Section IV. The goal of this controller is to expand the ROA by a desired amount while minimizing the battery effort. Lastly, the conclusion and future work are stated.

Throughout this paper, the following notation will be used. A matrix $H \in \mathbb{R}^{n \times n}$ is said to be positive definite (denoted as $H > 0$), if for any nonzero vector $x \in \mathbb{R}^n$ it satisfies $x^T H x > 0$. The set $\mathbb{R}_+$ is used to denote the set of all positive real numbers greater than 0. The set $\mathbb{N} = \{1, 2, \dots\}$ contains the natural numbers. Lastly, the symbol $\triangleq$ is used to denote equality by definition.

II. PROBLEM STATEMENT BASED ON ONE CPL

A. Problem Formulation

In order to visualize the problem, the microgrid shown in Fig. 1 is first studied with only one CPL and a dc source as shown in Fig. 2. The dynamic equations for this system are given by

$$\begin{aligned}\dot{x}_1 &= -\frac{r}{L}x_1 - \frac{1}{L}x_2 + \frac{1}{L}V_{dc} \\ \dot{x}_2 &= \frac{1}{C}x_1 - \frac{1}{C}\frac{P}{x_2}\end{aligned}\quad (1)$$

where $x_1 = i_L$ and $x_2 = v_C$ are the inductor current and capacitor voltage respectively and $r \triangleq r_{dc} + r_L$. In addition, we can see the nonlinear term P/x_2 . In this paper, the load power P is assumed constant.

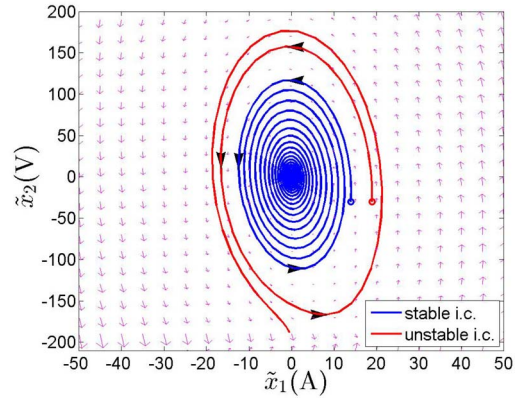


Fig. 3. System response and phase diagram of (1) based on two different initial conditions.

TABLE I
CIRCUIT PARAMETERS AND INITIAL CONDITIONS FOR FIG. 2

r	1.1 Ω	L	39.5 mH
C	500 μF	V_{dc}	200 V
P	300 W	x_0	(1.5, 198.34)
Stable i.c.	(14, -30)	Unstable i.c.	(19, -30)

It is possible to find a maximum value the load can demand by linearizing (1) about the equilibrium point, $x_o \triangleq (x_{10}, x_{20})^T$, to obtain

$$P < \min \left\{ \frac{V_{dc}}{4r}, \frac{rCx_{20}^2}{L} \right\} \triangleq P_{\max}. \quad (2)$$

The first term is due to finding a real-operating point and the second guarantees the eigenvalues of the Jacobian matrix have negative real part [6]. Due to the nonlinear nature of the load, a safe operation of these systems is not ensured by guaranteeing local stability. To observe this, a change of coordinates about the operating point, x_o , is performed as $\tilde{x} = x - x_o$. Under the new coordinate system, (1) becomes

$$\begin{aligned}\dot{\tilde{x}}_1 &= -\frac{r}{L}\tilde{x}_1 - \frac{1}{L}\tilde{x}_2 \\ \dot{\tilde{x}}_2 &= \frac{1}{C}\tilde{x}_1 + \frac{P}{Cx_{20}}\frac{\tilde{x}_2}{\tilde{x}_2 + x_{20}}.\end{aligned}\quad (3)$$

Fig. 3 shows the response of the system based on two different initial conditions with the parameters shown in Table I. It can be seen that only a difference in 5 A in the initial conditions caused the system from converging to the equilibrium to going unstable. Therefore, it is very important for the applications considered (e.g., aircraft, ships, etc.) to compute a ROA based on the system parameters.

B. Region of Attraction Review for One CPL

The method presented in [1] to compute the ROA is discussed briefly in the remainder of this section to show the importance of the ROA. The system in (1) is formulated as a T-S Fuzzy model. For a case with $Q \in \mathbb{N}$ CPLs, the T-S model will contain 2^Q Fuzzy rules and linear systems. Hence, for $Q = 1$, two linear systems defined as $\dot{x} = A_i x$, $i = 1, 2$ are

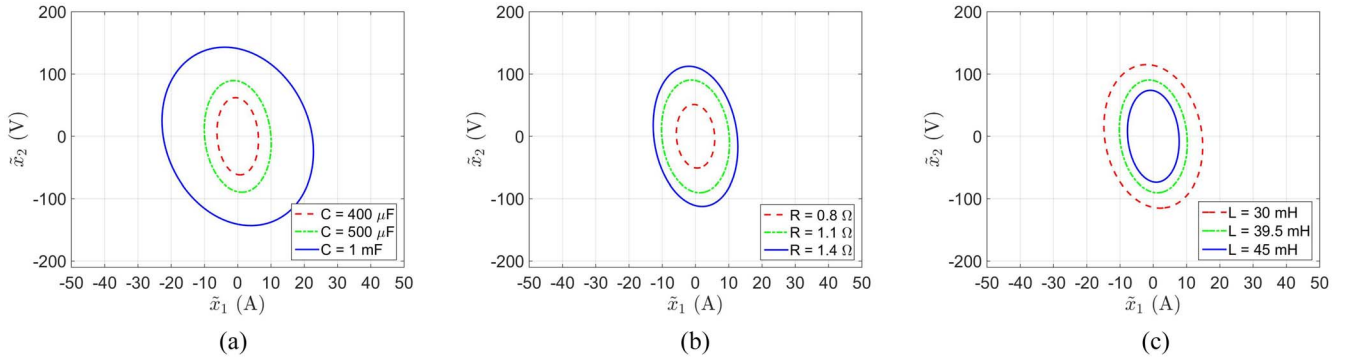


Fig. 4. Estimated ROAs based on different passive component values deviating from the nominal points in Table I. Estimated ROA based on different (a) capacitor values, (b) resistor values, and (c) inductor values.

Algorithm 1 Algorithm to Estimate ROA for $Q = 1$ [1]

```

1: procedure TSROA( $A_i, \delta$ )
2:   Set  $x_{2N\text{-new}} = \delta$ 
3:   Set  $x_{2N\text{-prev}} = 0$ 
4:   Compute  $A_i$  for  $i = 1, 2$ 
5:   Check (6)
6:   while (6) is feasible do
7:     Set  $x_{2N\text{-prev}} = x_{2N\text{-new}}$ 
8:     Enlarge  $\mathcal{D}_\delta$  by  $x_{2N\text{-new}} = x_{2N\text{-prev}} + \delta$ 
9:     Re-compute  $A_i$  for  $i = 1, 2$ 
10:    Check (6)
11:   end while
12:   Estimated ROA is computed using  $x_{2N\text{-prev}}$ 
13: end procedure

```

obtained by approximating the nonlinearity in a neighborhood of $\tilde{x}_2$, i.e., the Fuzzy model is valid for $\tilde{x}_2 \in [-x_{2N} \ x_{2N}] \triangleq \mathcal{D}_\delta$ and $x_{2N} > 0$ is a nominal value defining this set. The matrices are computed as

$$A_1 = \begin{pmatrix} \frac{-r}{L} & \frac{-1}{L} \\ \frac{1}{C} & \tilde{f}(x_{2N}) \end{pmatrix}, \quad A_2 = \begin{pmatrix} \frac{-r}{L} & \frac{-1}{L} \\ \frac{1}{C} & \tilde{f}(-x_{2N}) \end{pmatrix} \quad (4)$$

where

$$\tilde{f}(\tilde{x}_2) \triangleq \frac{P}{Cx_{20}\tilde{x}_2 + x_{20}}. \quad (5)$$

Thus, the system with one CPL is stable in the region defined by $\mathcal{D}_\delta$ if the following linear matrix inequality (LMI) is feasible:

$$W > 0 \\ A_i^T W + W A_i < 0 \text{ for } i = 1, 2. \quad (6)$$

To compute the ROA, we can iteratively increase $\mathcal{D}_\delta$ by a small positive number $\delta > 0$ until (6) is no longer feasible. This algorithm is described in Algorithm 1.

The estimated domain of attraction is then contained in the set $\mathcal{D} = \{x \in \mathbb{R}^n \mid |b^T x| < r = x_{2N\text{-prev}}\}$, where $b = (0, 1)^T$. Therefore, the largest level set of the Lyapunov equation $V = x^T W x$ contained in $\mathcal{D}$ can be computed by calculating

$$\min_{s.t. |b^T x| = r} x^T W x = \frac{r^2}{b^T W^{-1} b} \triangleq a \quad (7)$$

and the estimated ROA is $\Omega_a = \{x \in \mathbb{R}^2 \mid x^T W x < a\}$. As an example, ROAs based on different capacitor, resistor, and inductor values are shown in Fig. 4. It can be seen that a larger capacitance and resistance increases the robustness of the system whereas the inductance has an opposite effect.

However, there are several drawbacks of this method such as the need to iteratively increase the nonlinear region by a factor $\delta \in \mathbb{R}_+$ and the need to derive an exponential number of models based on the total CPLs. For a very large number of CPLs (Q), it is computationally expensive to use this method since at each step the 2^Q matrices (A_i) need to be recomputed as well as the $2^Q + 1$ LMIs.

III. PROPOSED FORMULATION

A. Single CPL

To overcome the computational drawbacks of the previous methods, (1) is approached from a different perspective. The system in (3) is first rewritten as a Lur'e problem [16] of the form

$$\dot{\tilde{x}}(t) = A\tilde{x}(t) + Dh(\tilde{x}) \quad (8)$$

where it is desired for the nonlinearity to satisfy a quadratic bound as

$$h^T(\tilde{x})h(\tilde{x}) \leq \alpha^2 \tilde{x}^T H^T H \tilde{x} \quad (9)$$

where $\alpha \in \mathbb{R}_+$. The matrices A , D , and $H \geq 0$ are defined as

$$A = \begin{pmatrix} \frac{-r}{L} & \frac{-1}{L} \\ \frac{1}{C} & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 \\ \frac{1}{C} \end{pmatrix}, \quad H = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (10)$$

and the nonlinear term is

$$h(\tilde{x}) = \frac{P\tilde{x}_2}{x_{20}(\tilde{x}_2 + x_{20})}. \quad (11)$$

The goal is to guarantee asymptotic stability of the system whenever the inequality (9) is satisfied for a certain $\alpha > 0$. This goal can be achieved if for an $\alpha > 0$, it is possible to find a $W > 0$ such that the following holds:

$$V(\tilde{x}) = \tilde{x}^T W \tilde{x} > 0 \quad (12)$$

$$\dot{V}(\tilde{x}) = \tilde{x}^T (A^T W + W A) \tilde{x} + h^T D^T W \tilde{x} + \tilde{x}^T W D h < 0$$

$$\text{whenever } h^T(\tilde{x})h(\tilde{x}) \leq \alpha^2 \tilde{x}^T H^T H \tilde{x}. \quad (13)$$

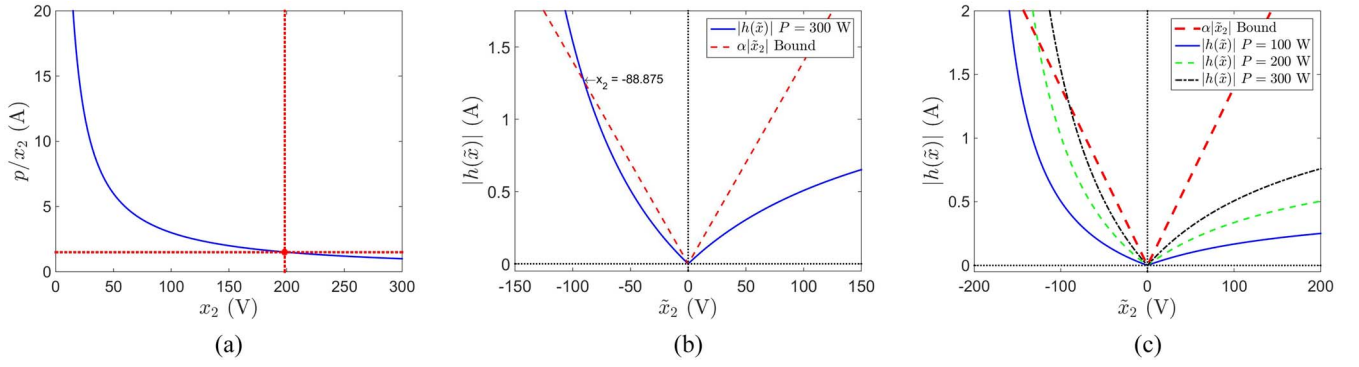


Fig. 5. Illustration on computing bound α using (17) for one CPL. (a) CPL characteristics for $P = 300$ W, the red circle shows the operating point. (b) Nonlinearity and computed bound in new coordinate system. (c) Intersection points for different load powers.

Using LMIs, (12) and (13) can be rewritten as

$$\begin{pmatrix} A^T W + WA & WD \\ D^T W & 0 \end{pmatrix} < 0 \quad \text{whenever} \quad W > 0 \quad (14)$$

$$\begin{pmatrix} \alpha^2 H^T H & 0 \\ 0 & -I \end{pmatrix} \geq 0. \quad (15)$$

The previous equations can be simplified further using s-procedure and Schur's complement [17]

$$\begin{pmatrix} YA^T + AY & D & YH^T \\ D^T & -I & 0 \\ HY & 0 & -\gamma I \end{pmatrix} < 0 \quad \text{whenever} \quad Y > 0 \quad (16)$$

where $\gamma = (1/\alpha^2)$ and $Y = \tau W^{-1}$ for some $\tau > 0$ (s-procedure constant).

An SDP problem can be formulated to maximize α to enlarge the ROA (equivalent to minimizing γ). SDP is a specific class of convex optimization problems with linear cost function and LMI constraints [18]. Therefore, using (16), the following SDP is setup to maximize α and guarantee asymptotic stability whenever (9) holds:

$$\begin{aligned} \min \quad & \gamma \\ \text{subject to:} \quad & Y > 0 \\ & \begin{pmatrix} AY + YA^T & D & YH^T \\ D^T & -I & 0 \\ HY & 0 & -\gamma I \end{pmatrix} < 0. \end{aligned} \quad (17)$$

Problem (17) is derived using a similar procedure as described in [19]. Using the CVX toolbox [18], we obtained a maximum $\alpha = 0.0138$ for the parameters in Table I. Note that for the case with one CPL, the inequality in (9) implies $|h(x)| \leq \alpha|\tilde{x}_2|$. Thus, it is possible to compute the intersection of $|h(x)|$ and $\alpha|\tilde{x}_2|$ to calculate at which value of the capacitor voltage this inequality will be violated [the ROA can then be calculated using this value and W from (17)]. A similar ROA is obtained using this procedure as the ones shown using the previous T-S Fuzzy method. This method and its results for one CPL are outlined in Fig. 5. In addition, Fig. 5(c) shows how the intersection point (and thus the ROA) decreases as the load power increases.

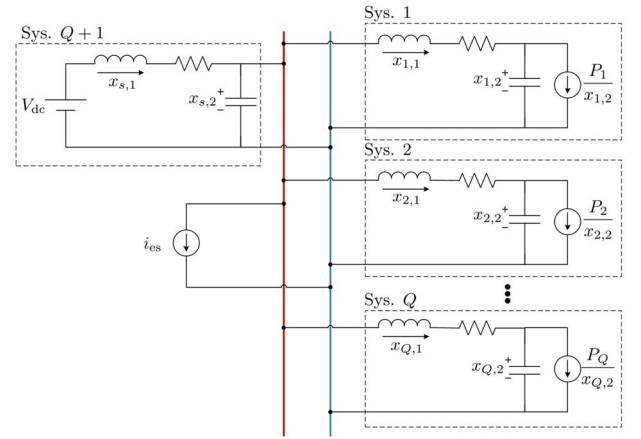


Fig. 6. Simplified version of the dc microgrid shown in Fig. 1 with Q CPLs.

B. Multiple CPLs

The proposed method to compute the ROA will be expanded to include Q CPLs and an energy storage connected to the source as shown in Fig. 6. To model this large dynamic system, we can decouple it into $Q+1$ subsystems. The CPLs $(1, \dots, Q)$ are of the form

$$\dot{x}_j = A_j x_j + d_j h_j + A_{js} x_s \quad (18)$$

for $j \in \mathcal{J}_Q \triangleq \{1, 2, \dots, Q\}$ and $s \triangleq Q+1$ represents the source. The matrices A_j , d_j , and A_{js} are of the following form:

$$A_j = \begin{pmatrix} -r_j/L_j & -1/L_j \\ 1/C_j & 0 \end{pmatrix}, \quad d_j = \begin{pmatrix} 0 \\ 1/C_j \end{pmatrix}, \quad A_{js} = \begin{pmatrix} 0 & 1/L_j \\ 0 & 0 \end{pmatrix}. \quad (19)$$

The source subsystem can be written as

$$\dot{x}_s = A_s x_s + b_s V_{dc} + b_{es} i_{es} + \sum_{j=1}^Q A_{cnj} x_j \quad (20)$$

with matrices

$$A_s = \begin{pmatrix} -r_s/L_s & -1/L_s \\ 1/C_s & 0 \end{pmatrix}, \quad b_s = \begin{pmatrix} 1/L_s \\ 0 \end{pmatrix}, \quad A_{cn} = \begin{pmatrix} 0 & 0 \\ -1/C_s & 0 \end{pmatrix}, \quad b_{es} = \begin{pmatrix} 0 \\ -1/C_s \end{pmatrix}. \quad (21)$$

TABLE II
PARAMETERS FOR DC MICROGRID WITH $Q = 2$ CPLs

r_1	0.8 Ω	r_2	0.42 Ω	r_3	0.4 Ω
L_1	40 mH	L_2	19.6 mH	L_3	17.3 mH
C_1	1 mF	C_2	1.05 mF	C_3	1.05 mF
P_1	900 W	P_2	600 W	V_{dc}	200 V

Assuming a coordinate change about an operating point as $\tilde{x} = x - x_o$ and let the energy storage current i_{es} be the control input, we can rewrite the dc microgrid as a dynamic system of the form

$$\dot{\tilde{x}} = A\tilde{x} + Dh + B_{es}\tilde{i}_{es} + B_s\tilde{V}_{dc} \quad (22)$$

where the matrices A , D , B_{es} , and B_s are given in the appendix. The state variables are defined as $x = (x_1, \dots, x_Q, x_s)^T$ and the nonlinear term $h(\tilde{x}) = (h_1, \dots, h_Q)^T$ where

$$h_j(\tilde{x}_j) = \frac{P_j \tilde{x}_{j,2}}{x_{j,20}(\tilde{x}_{j,2} + x_{j,20})}, \quad j \in \mathcal{J}_Q. \quad (23)$$

The term $\tilde{x}_{j,2}$ is used to denote the second state (capacitor voltage) of the j th CPL subsystem and $x_{j,20}$ is the corresponding state operating point.

In this case, we assume the nonlinear term must satisfy a quadratic bound of the form

$$h^T(x)h(x) = \sum_{j=1}^Q h_j^T h_j \leq \alpha^2 \tilde{x}^T H^T H \tilde{x} \quad (24)$$

for a diagonal matrix $H \geq 0$. Next, we consider two cases.

1) *Robustness Index*: The parameter α can be seen as a robustness index for the dc microgrid since the larger this number is, the more robust the system is to nonlinear disturbances. As a first case, we consider a matrix H of the form

$$H = \begin{pmatrix} H_1 & \dots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & H_Q & 0 \\ 0 & \dots & 0 & 0 \end{pmatrix}, \quad H_i = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (25)$$

for $i \in \mathcal{J}_Q$. With the system in (22) and H as shown in the previous equation, we can setup a problem to maximize α similar to (17). This procedure is then applied to test a case with $Q = 2$ CPLs (six states). The parameters of a dc microgrid with two CPLs are shown in Table II. Based on these values, the nominal robustness index obtained using (17) is $\alpha = 0.0224$.

2) *Region of Attraction*: The previous case can be used to compute a maximum bounding parameter α based on a given H . However, it is still important to compute the ROA even for a system with multiple CPLs. From this point on, we let $x = \tilde{x}$ for simplicity. Assuming $x \in \mathbb{R}^n$, consider a region of the form

$$\Lambda(a) = \{x \in \mathbb{R}^n \mid |x_{j,2}| \leq a, \quad a > 0 \quad j \in \mathcal{J}_Q\}. \quad (26)$$

Next, assume there exists a diagonal matrix $F \geq 0$ such that

$$h^T(x)h(x) \leq \alpha^2 x^T F^T F x \quad (27)$$

Algorithm 2 Algorithm to Maximize $\Lambda(a)$ for Arbitrary Q

```

1: procedure SDPCPL( $A, D, a, \delta$ )
2:   compute  $F = \Phi^{\frac{1}{2}}(a)$ 
3:   set  $H = F$ 
4:   set  $a_{\text{prev}} = 0$ 
5:   solve (17)
6:   while  $\alpha \geq 1$  do
7:      $a_{\text{prev}} = a$ 
8:      $a = a_{\text{prev}} + \delta$ 
9:     re-compute  $F = \Phi^{\frac{1}{2}}(a)$ 
10:    set  $H = F$ 
11:    solve (17)
12:   end while
13:   ROA is given by largest set  $\Pi(r) \subset \Lambda(a_{\text{prev}})$ 
14: end procedure

```

for all $x \in \Lambda(a)$ and a positive constant $\alpha \geq 1$. If problem (17) is feasible with $H = F$ and $\alpha \geq 1$, then the estimated ROA can be computed as the largest set

$$\Pi(r) = \{x \in \mathbb{R}^n \mid V(x) = x^T Y^{-1} x \leq r\} \quad (28)$$

contained in $\Lambda(a)$ [20].

To compute F , we can exploit the advantages of the nonlinear terms in (23) which can be written of the form

$$h_j(x_j) = \frac{P_j}{x_{j,20}(x_{j,2} + x_{j,20})} \cdot x_{j,2} \triangleq m_j(x) x_{j,2} \quad (29)$$

for $x_{j,20} > 0$ and $j \in \mathcal{J}_Q$. Thus

$$h^T(x)h(x) = x^T M(x)x \quad (30)$$

where $M(x)$ has diagonal terms consisting of the functions $m_j^2(x)$. It is then possible to find a matrix $\Phi(x)$ with diagonal entries $\phi_j^2(x)$ such that

$$x^T M(x)x \leq x^T \Phi(x)x, \quad \forall x \text{ s.t. } \phi_j(x) > 0, \quad j \in \mathcal{J}_Q. \quad (31)$$

The matrices $M(x)$, $\Phi(x)$, and the functions $\phi_j(x)$ are shown in the appendix. Let $\mathcal{T} = \{x \in \mathbb{R}^n \mid \phi_j(x) > 0 \quad j \in \mathcal{J}_Q\}$. For a positive number a satisfying $\Lambda(a) \subset \mathcal{T}$, the matrix $F = \Phi^{(1/2)}(a)$ meets the desired requirement

$$h^T(x)h(x) \leq x^T F^T F x \quad \forall x \in \Lambda(a). \quad (32)$$

An algorithm to enlarge the set $\Lambda(a)$ and compute the ROA is then shown in Algorithm 2. An initial value of a should be chosen to satisfy $\alpha > 1$ from (17). Fig. 7 shows several iterations of the proposed algorithm to expand $\Lambda(a)$ based on the parameters shown in Table I (one CPL). Furthermore, using this method a similar region is computed as the ones obtained by calculating the intersection point shown in Section III-A and the T-S method.

The advantage of the proposed formulation can be seen for computing the ROA for a dc microgrid multiple CPLs. Using this method only one SDP is needed to be solved for each iteration, the number of LMIs do not increase exponentially with Q , and only the bounding matrix F is recomputed each iteration. Using Algorithm 2 and the values in Table II (system

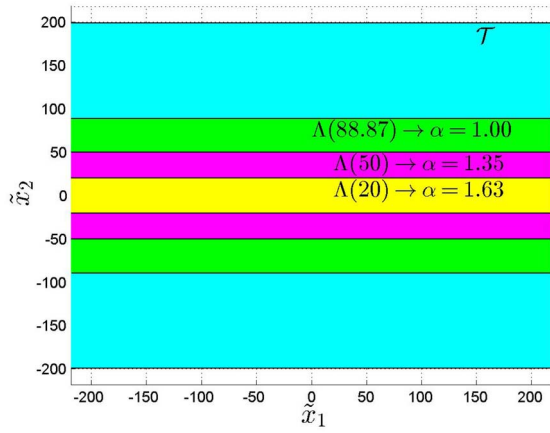


Fig. 7. Plot showing the expansion of the set $\Lambda(a)$ using Algorithm 2 for one CPL using the nominal parameters in Table I.

with two CPLs), a maximum constraint set $\Lambda(22.6)$ was found for which the ROA can then be computed. Since we have a six dimensional system, we illustrate the ROA for this system by its intersection with different planes as shown in Fig. 8 (without control/dashed blue curves).

IV. CONTROLLER DESIGN

A. Feedback Design

Energy storage can be used to increase the robustness of the system by mitigating the transients which occur due to temporary faults at the dc link or disturbances in the dc source voltage. Therefore, it is possible to increase the ROA using the proper choice of feedback. In this section, we consider the use of a linear controller to achieve this goal by a desired amount while minimizing the norm of the feedback gain.

The method to compute the ROA presented in the last section can be extended to calculate a feedback of the form $i_{es} = K\tilde{x}$ (using the energy storage input) to increase this region. For example, the nominal robustness index of the network shown in Table II was computed as $\alpha_n = 0.0224$. Assuming that we would like to increase this index to $\alpha = \bar{\alpha} > \alpha_n$ by optimally computing K , we can setup the following SDP problem [19]:

$$\begin{aligned}
 \min \quad & b\gamma + b_L\kappa_L + b_Y\kappa_Y \\
 \text{subject to} \quad & Y \succ 0 \\
 & \begin{pmatrix} AY + YA^T + B_{es}L + L^TB_{es}^T & D & YH^T \\ D^T & -I & 0 \\ HY & 0 & -\gamma I \end{pmatrix} \prec 0 \\
 & \begin{pmatrix} -\kappa_L I & L^T \\ L & -I \end{pmatrix} \prec 0, \quad \begin{pmatrix} Y & I \\ I & \kappa_Y I \end{pmatrix} \succ 0 \\
 & \gamma - \frac{1}{\bar{\alpha}^2} < 0
 \end{aligned} \tag{33}$$

where the feedback matrix can be calculated as: $K = LY^{-1}$, the constants b , b_L , and b_Y are used as weighting factors, and the variables κ_L , κ_Y bound the norm of the feedback matrix as

$$\|K\| \leq \|L\| \|Y^{-1}\| < \sqrt{\kappa_L\kappa_Y}. \tag{34}$$

Comparing (33) to (17), the new LMIs in (33) can be derived from (34) as follows:

$$\begin{aligned}
 \|L\| < \sqrt{\kappa_L}, \quad & \Leftrightarrow L^TL - \kappa_L I \prec 0 \\
 \Rightarrow \begin{pmatrix} -\kappa_L I & L^T \\ L & -I \end{pmatrix} \prec 0 \quad & \text{(Schur's complement)} \tag{35}
 \end{aligned}$$

and

$$\begin{aligned}
 \begin{pmatrix} Y & I \\ I & \kappa_Y I \end{pmatrix} \succ 0 \quad & \Rightarrow \kappa_Y I - Y^{-1} \succ 0 \\
 \Rightarrow \|Y^{-1}\| < \kappa_Y. \quad & \tag{36}
 \end{aligned}$$

Hence (34) is satisfied [19]. The last term, $\gamma - (1/\bar{\alpha}^2) < 0$, makes sure the desired robustness index, $\bar{\alpha}$, can be achieved.

It is important to note that the system in (22) with the energy storage as an input does not satisfy the matching conditions described in [19] and [21]. These conditions state that if both the control input (i_{es}) and the nonlinear disturbance ($h(x)$) enter the system through the same matrix, e.g., B , then the system can be robustly stabilized with arbitrarily large index α by a (possibly large) proper choice of feedback K [19, Th. 3.5]. In (22), the input and the disturbance enter through different matrices, B_{es} and D respectively, and $\text{Im}D \not\subset \text{Im}B_{es}$ (if $\text{Im}D \subset \text{Im}B_{es}$, then $D = B_{es}C$ for some matrix C , and the matching conditions would be satisfied). Therefore, the system can only be improved by a limited amount. However, if energy storage or current source inputs were to be considered in parallel to each CPL, as is the case in [12], then the matching conditions could be met and the system can be robustly stabilized with arbitrarily large index α using a simpler version of (33), where the norm of the feedback matrix K is not to be bounded. Nevertheless, for the problem considered in this paper of using only one energy storage, the increase in robustness can be sufficient for even severe fault scenarios.

B. Case Study

The main structure of the controller is shown in Fig. 9. Simulations were then conducted to increase the robustness index of the microgrid in Table II to $\alpha = 0.045 > \alpha_n$. The computed feedback gain using (33) is $K = (1.406 \ -0.116 \ 0.47 \ 0.354 \ 0.653 \ 0.121)$. To demonstrate the advantages of this controller, we assume the system is working with the nominal parameters shown in Table II and introduce a fault at the dc link for 2 ms. This fault causes the system to go unstable without any control, whereas the response of the system with the proposed controller is shown in Fig. 10. It can be seen that the system with control is able to return to its operating point after the fault. Since the robustness index was increased, the ROA for the system with the proposed controller is also increased. A comparison of the ROAs with and without control is shown in Fig. 8. The solid red curves represent the ROA with the proposed controller whereas the dashed blue curves shows the ROA without any control. It can be seen that a significant enlargement of the ROA was achieved by increasing the robustness index while at the same time minimizing the norm of the feedback matrix K (amount of current supplied by battery).

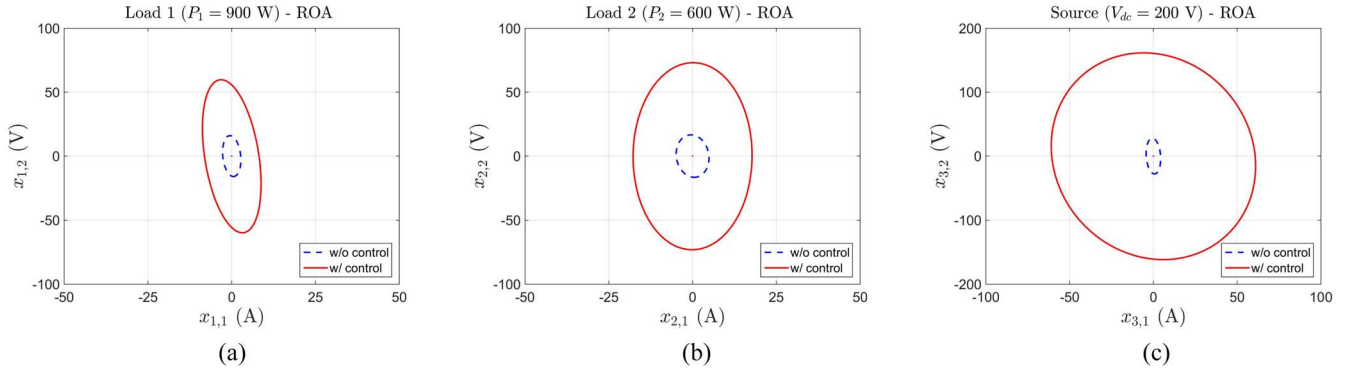


Fig. 8. Comparison of the ROAs for two CPLs using Algorithm 2 with and without proposed controller. Intersection for the estimated ROA with the plane given by (a) $x_2 = x_3 = (0, 0)^T$, (b) $x_1 = x_3 = (0, 0)^T$, and (c) $x_1 = x_2 = (0, 0)^T$.

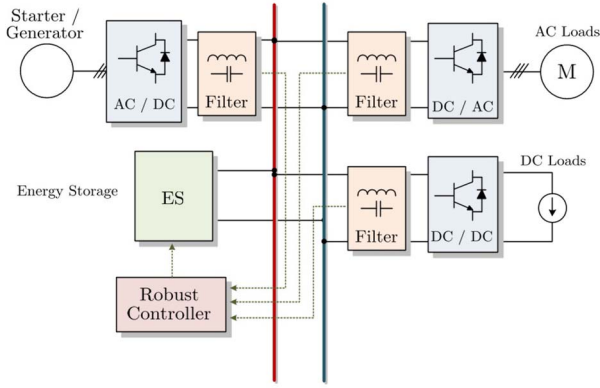


Fig. 9. Robust controller using energy storage to increase robustness margin.

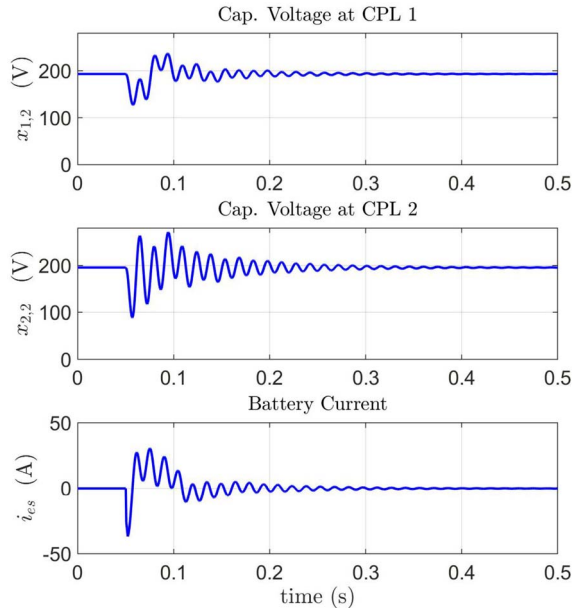


Fig. 10. Results of the closed loop system to a short circuit fault in the dc link at $t = 0.05$ s. The feedback gain K was computed using (33) to increase the robustness index (hence ROA) to $\alpha = 0.045$.

V. CONCLUSION

In this paper, a new methodology to analyze the stability of a dc microgrids composed of multiple CPLs is proposed.

The system was formulated as a Lur'e problem with quadratic bounds. An SDP problem was formulated to compute a robustness index and an algorithm using the same SDP problem was presented to calculate an estimate of the domain of attraction. The main advantages of this method is that its complexity does not increase exponentially with the number of CPLs. The use of energy storage was studied to mitigate temporary faults such as a short circuit in the dc link or disturbances in the dc source voltage. With a similar SDP formulation, it is shown how to compute the feedback matrix K to achieve a desired robustness index while minimizing the norm of K . Future work aims at validating the proposed strategies with Hardware in the Loop simulations and stabilization of the case where the load power varies within a predefined set.

APPENDIX

The matrices for the large scale dc microgrid model shown in (22) are of the form

$$A = \begin{pmatrix} A_1 & \cdots & 0 & A_{1s} \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & A_Q & A_{Qs} \\ A_{cn} & \cdots & A_{cn} & A_s \end{pmatrix}, \quad D = \begin{pmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_Q \\ 0 & \cdots & 0 \end{pmatrix}$$

$$B_{es} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ b_{es} \end{pmatrix}, \quad \text{and} \quad B_s = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ b_s \end{pmatrix}. \quad (37)$$

The matrices $M(x)$ used to compute $h^T(x)h(x)$ is given as

$$M(x) = \begin{pmatrix} M_1 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & M_Q & 0 \\ 0 & \cdots & 0 & 0 \end{pmatrix}, \quad M_j = \begin{pmatrix} 0 & 0 \\ 0 & m_j^2(x) \end{pmatrix} \quad (38)$$

where the terms $m_j(x)$ are shown in (29) and $j = 1, \dots, Q$. Lastly, the matrix $\Phi(x)$ used to compute F is given as

$$\Phi(x) = \begin{pmatrix} \Phi_1 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & \Phi_Q & 0 \\ 0 & \cdots & 0 & 0 \end{pmatrix}, \quad \Phi_j = \begin{pmatrix} 0 & 0 \\ 0 & \phi_j^2(x) \end{pmatrix} \quad (39)$$

where

$$\phi_j(x) = \frac{P_j}{x_{j,20}(-|x_{j,2}| + x_{j,20})} \quad (40)$$

for $j = 1, \dots, Q$.

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